

SGA 5 — Exposé I

Dualizing Complexes

4. Local duality

The object of this section is to present the notion of “biduality” developed in the preceding sections in a form closer to geometric intuition.

4.1

Let X be a prescheme, let $i : x \rightarrow X$ be a closed point whose residue field is separably closed, and let K_X be a dualizing complex on X (1.7). We know (1.15) that $K_{\{x\}} = R^!i(K_X)$ is a dualizing complex on x , hence, by (2.1) and (3.4.4), isomorphic to $A[r]$ for some $r \in \mathbb{Z}$ (if A is local). We shall say that K_X is *normalized at x* if $r = 0$ and if an isomorphism $K_{\{x\}} \xrightarrow{\sim} A$ has been given.

For example, if X is smooth of relative dimension d over the spectrum of a separably closed field, the complex $(\mu_n)_X^{\otimes d}[2d]$ is, by the purity theorem (SGA XVI 3, SGA 4^{1/2} Cycle 2.3), canonically normalized at every closed point of X .

4.2

Let X be a prescheme, $i : Y \rightarrow X$ a closed sub-prescheme, and K_X a dualizing complex on X . Then (1.15) $K_Y = R^!i(K_X)$ is a dualizing complex on Y , and one has the *complementary induction formula* (1.12 b) (ii)), for $F \in \text{ob } D_c^b(X)$,

$$(4.2.1) \quad i^* \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y R^!i(F),$$

where $\underline{D}_X = R \underline{\text{Hom}}(\ , K_X)$ and $\underline{D}_Y = R \underline{\text{Hom}}(\ , K_Y)$.

Apply (4.2.1) in particular in the case where $Y = \{x\}$ is a closed point of X with separably closed residue field, K_X being normalized at x (4.1). Taking into account that A_x is injective, (4.2.1) is then written

$$(4.2.2) \quad \underline{D}_X(F)_x \xrightarrow{\sim} \text{Hom}^*(R^!i(F), A).$$

Noting that $R^!i(F)$ is also written, by definition, $R\Gamma_x(F)$, one therefore has a perfect pairing in $D(A)$:

$$(4.2.2)' \quad \underline{D}_X(F)_x \times R\Gamma_x(F) \longrightarrow A,$$

giving rise to perfect pairings between finite groups:

$$(4.2.2)'' \quad \underline{H}^i(\underline{D}_X(F))_x \times \underline{H}_x^{-i}(F) \longrightarrow A.$$

For example, let X be a prescheme smooth of dimension d over a separably closed field, and let $x \in X$ be a closed point. Then, for $F \in \text{ob } D_c^b(X)$, one has a canonical perfect duality between finite groups:

$$(4.2.2)''' \quad \text{Ext}^{2d-i}(F, \mu_n^{\otimes d})_x \times \underline{H}_x^i(F) \longrightarrow A.$$

Exercise 4.2.3

Let X be a strictly local scheme (SGA VIII 4.2), with closed point x , and let $i : x \rightarrow X$ be the canonical morphism. Let K be a dualizing complex on X , normalized at x . Denote by

$$\mathbf{L} \otimes_{i?} : D^b(A) \longrightarrow D^-(X)$$

the functor defined by

$$\mathbf{L} \otimes_{i?}(M) = M \otimes_A^{\mathbf{L}} K.$$

(If K has finite tor-dimension over A , $\mathbf{L} \otimes_{i?}$ extends to a functor from $D^+(A)$ to $D(X)$.)

Show that there exists, for $L \in D^-(X)$ and $M \in D^b(A)$, a canonical functorial isomorphism

$$(*) \quad \mathrm{R} \mathrm{Hom}(L, \mathbf{L} \otimes_{i?}(M)) \xrightarrow{\sim} \mathrm{R} \mathrm{Hom}(\mathrm{R}^! i(L), M),$$

(assuming of course that $\mathrm{R}^! i : D^-(X) \rightarrow D^-(A)$ is defined ...).

Hint: First define a “trace” morphism $\mathrm{R}^! i \otimes_{i?}^{\mathbf{L}}(M) \rightarrow M$; then, to check that $(*)$ is an isomorphism, reduce by dévissage to the case where $L \in \mathrm{ob} D_c^b(X)$ and $M = A$, and apply (4.2.2). Note that, for $M = A$, $(*)$ reduces to the perfect pairing (“local duality”, cf. [H] V 6.2)

$$\mathrm{R} \Gamma_x(L) \times \mathrm{R} \mathrm{Hom}(L, K) \longrightarrow A,$$

or

$$H_x^i(L) \times \mathrm{Ext}^{-i}(L, K) \longrightarrow A.$$

4.3

We shall now generalize (4.2.2) to the case of a geometric point $\bar{x}$ of X localized at a point x which is not necessarily closed. For this we need a few preliminaries.

Lemma 4.3. Let X be a prescheme, X' the strict localization of X at a geometric point $\bar{x}$, and $f : X' \rightarrow X$ the canonical map. Then, for $F \in \mathrm{ob} D_c^-(X)$ and $G \in \mathrm{ob} D^+(X)$, the canonical functorial homomorphism (SGA VI I 3.7)

$$f^* \mathrm{R} \underline{\mathrm{Hom}}(F, G) \longrightarrow \mathrm{R} \underline{\mathrm{Hom}}(f^* F, f^* G)$$

is an isomorphism.

Proof. As usual, one reduces to F and G concentrated in degree 0. Since the question is local in a neighborhood of x , one may suppose X affine. Then F admits (SGA IX 2.7) a left resolution by sheaves of the form $i_!(A_U)$, where $i : U \rightarrow X$ is étale of finite presentation; hence, by dévissage, one is reduced to $F = i_!(A_U)$. Form the cartesian square

$$\begin{array}{ccc} U & \xleftarrow{g} & U' \\ \uparrow i & & \uparrow i' \\ X & \xleftarrow{f} & X' \end{array}$$

One has canonical isomorphisms, trivial special cases of duality (SGA XVIII 3.1.10):

$$\mathrm{R}\underline{\mathrm{Hom}}(F, G) \xleftarrow{\sim} \mathrm{R}i_*i^*G, \quad \mathrm{R}\underline{\mathrm{Hom}}(f^*F, f^*G) \xleftarrow{\sim} \mathrm{R}i'_*g^*i^*G.$$

Under these isomorphisms, the canonical arrow

$$f^*\mathrm{R}\underline{\mathrm{Hom}}(F, G) \longrightarrow \mathrm{R}\underline{\mathrm{Hom}}(f^*F, f^*G)$$

is identified with the base-change arrow (SGA XVII)

$$f^*\mathrm{R}i_*i^*(G) \longrightarrow \mathrm{R}i'_*g^*i^*(G).$$

But this is an isomorphism, as one sees at once by passing to the limit (SGA VII 5.11).

Lemma 4.4. Let X be a prescheme, and let $K \in \mathrm{ob} D_c^+(X)$ be a complex satisfying conditions (i) and (ii) of (1.7). In order that K be dualizing, it is necessary and sufficient that, for every strict localization $f : \overline{X}(\overline{x}) \rightarrow X$, f^*K be dualizing.

Proof. Use the fact that every constructible sheaf on a strict localization $\overline{X}(\overline{x})$ is induced by a constructible sheaf on an étale $\overline{x}$ -pointed scheme over X (SGA IX 2.7.4), and apply (4.3).

4.5

Let X be a prescheme, $\overline{x}$ a geometric point of X , and $f : X' = \overline{X}(\overline{x}) \rightarrow X$ the corresponding strict localization.

Let Y be the integral closed sub-prescheme which is the closure of x in X ; denote by $g : \overline{x} \rightarrow Y$ and $i : Y \rightarrow X$ the canonical maps, and form the cartesian square

$$\begin{array}{ccc} Y & \xrightarrow{i} & X \\ g \uparrow & & \uparrow f \\ \overline{x} & \xrightarrow{i'} & X' \end{array}$$

($\overline{x}$ is both the closed point of X' and the strict localization of Y at $\overline{x}$).

We shall denote by

$$\mathrm{R}\Gamma_{\overline{x}} : D^+(X) \longrightarrow D^+(\overline{x})$$

the functor defined by

$$(4.5.1) \quad \mathrm{R}\Gamma_{\overline{x}} = g^*\mathrm{R}^!i.$$

Using the well-known relation between the functors $\mathrm{R}^!i$ and $\mathrm{R}j_*j^*$, where $j : U \rightarrow X$ is the open complement of Y , and applying passage to the limit (SGA VII 5.11), one obtains a canonical isomorphism

$$(4.5.2) \quad \mathrm{R}\Gamma_{\overline{x}} \xrightarrow{\sim} \mathrm{R}^!i'f^*.$$

When x is a closed point with separably closed residue field, the functor $\mathrm{R}\Gamma_{\overline{x}}$ coincides with the usual functor $\mathrm{R}\Gamma_x$ (SGA V 4.3), which justifies the notation.

Let $K_X \in \mathrm{ob} D^+(X)$, and put $K_Y = \mathrm{R}^!j(K_X)$, $K_{X'} = f^*K_X$, and $K_{\overline{x}} = \mathrm{R}\Gamma_{\overline{x}}(K_X)$ (warning: $K_{\overline{x}}$ is not the fiber of K_X at $\overline{x}$). Suppose that K_X is dualizing. Then, by (1.15) and (4.4), the same is true of $K_{X'}$, K_Y , and $K_{\overline{x}}$. Hence (2.1) one has $K_{\overline{x}} \simeq A[d]$ for some $d \in \mathbb{Z}$.

We shall say that K_X is *normalized at $\bar{x}$* if $d = 0$ and if an isomorphism $K_{\bar{x}} \simeq A$ has been chosen. In the case where x is closed with separably closed residue field, this recovers the notion introduced in (4.1).

Let K_X be a dualizing complex, and put $\underline{D}_X = \mathbf{R}\underline{\mathrm{Hom}}(\ , K_X)$, $\underline{D}_Y = \mathbf{R}\underline{\mathrm{Hom}}(\ , K_Y)$, etc. Let $F \in \mathrm{ob} \, \mathbf{D}_c^b(X)$; let us compute $(\underline{D}_X F)_{\bar{x}}$. One has:

$$\begin{aligned} (\underline{D}_X F)_{\bar{x}} &\xrightarrow{\sim} g^* i^* \underline{D}_X F \\ &\xrightarrow{\sim} g^* \underline{D}_Y \mathbf{R}^! i F \quad (\text{by the complementary induction formula (4.2.1)}) \\ &\xrightarrow{\sim} \underline{D}_{\bar{x}} g^* \mathbf{R}^! i(F) \quad (\text{by (4.3)}), \text{ that is,} \\ (4.5.3) \quad (\underline{D}_X F)_{\bar{x}} &\xrightarrow{\sim} \underline{D}_{\bar{x}} \mathbf{R}\Gamma_{\bar{x}}(F). \end{aligned}$$

Of course, one could have made an analogous computation by passing through X' instead of Y ; one would then have obtained a canonical isomorphism

$$(\underline{D}_X F)_{\bar{x}} \xrightarrow{\sim} \underline{D}_{\bar{x}} \mathbf{R}^! i' f^*(F),$$

compatible with (4.5.3) by means of the identification (4.5.2).

If moreover K_X is normalized at $\bar{x}$, one obtains perfect pairings generalizing (4.2.2)' and (4.2.2)'':

$$(4.5.3)' \quad \mathbf{H}_Y^i(F)_{\bar{x}} \times \mathbf{R}^{-i} \Gamma_{\bar{x}}(F) \longrightarrow A,$$

$$(4.5.3)'' \quad \mathbf{H}^i(\underline{D}_X(F))_{\bar{x}} \times \mathbf{H}_{\bar{x}}^{-i}(F) \longrightarrow A,$$

where $\mathbf{H}_{\bar{x}}^i = \mathbf{H}^i \mathbf{R}\Gamma_{\bar{x}}$.

Conversely, let $K_X \in \mathrm{ob} \, \mathbf{D}^+(X)$ satisfy conditions (i) and (ii) of (1.7), and suppose that $K_{\bar{x}} = \mathbf{R}\Gamma_{\bar{x}}(K_X)$ is dualizing for every geometric point $\bar{x}$ of X . Then, for every $\bar{x}$, one can define (cf. 1.12 b) (ii)) a canonical arrow

$$(4.5.3)^{\mathrm{bis}} \quad \underline{D}_X(F)_{\bar{x}} \longrightarrow \underline{D}_{\bar{x}} \mathbf{R}\Gamma_{\bar{x}}(F) \quad (F \in \mathrm{ob} \, \mathbf{D}_c^b(X)).$$

If this is an isomorphism for every $\bar{x}$ and every F , then K_X is dualizing. Indeed, it remains to verify that the canonical arrow

$$F \longrightarrow \underline{D}_X \underline{D}_X(F)$$

is an isomorphism for every $F \in \mathrm{ob} \, \mathbf{D}_c^b(X)$. At an arbitrary geometric point $\bar{x}$, one has

$$\begin{aligned} F_{\bar{x}} &\xrightarrow{\sim} \underline{D}_{\bar{x}} \underline{D}_{\bar{x}}(F_{\bar{x}}) \quad (\text{since } K_{\bar{x}} \text{ is dualizing}) \\ &\xrightarrow{\sim} \underline{D}_{\bar{x}} \mathbf{R}\Gamma_{\bar{x}} \underline{D}_X(F) \quad (\text{by (4.3) and the induction formula 1.12 b) (i)}) \\ &\xrightarrow{\sim} \underline{D}_{\bar{x}} \underline{D}_{\bar{x}}(F)_{\bar{x}} \quad (\text{by hypothesis}), \end{aligned}$$

and therefore, modulo a compatibility check, (F, K_X) is bidualizing, which proves the assertion. In summary:

Proposition 4.5.4. Let X be a prescheme, and let $K_X \in \mathrm{ob} \, \mathbf{D}^+(X)$ be a complex satisfying conditions (i) and (ii) of (1.7). In order that K_X be dualizing, it is necessary and sufficient that, for every geometric point $\bar{x}$ of X , the complex $K_{\bar{x}}$ be dualizing and the canonical arrow $(4.5.3)^{\mathrm{bis}}$ be an isomorphism for every $F \in \mathrm{ob} \, \mathbf{D}_c^b(X)$.

Remark 4.5.5. The existence of dualizing complexes should be regarded, in a sense, as an important complement to the global duality theorem. Indeed, the latter expresses that, for a separated morphism of finite type $f : X \rightarrow Y$ (Y quasi-compact), one has a canonical isomorphism

$$\underline{D}_Y \mathrm{R}_! f(F) \xrightarrow{\sim} \mathrm{R}f_* \underline{D}_X(F)$$

(with the notation of 1.12). Its use for computing $\mathrm{R}_! f(F)$ (when $F \in \mathrm{ob} \mathrm{D}_c^b(X)$) requires information about $\underline{D}_X(F)$. When K_X is dualizing, this information is supplied by the isomorphism (4.5.3), which makes possible, at least in theory, a computation of the fibers of $\underline{D}_X(F)$ as invariants of “purely spatial” cohomology.

Exercise 4.5.6. Let X be a prescheme endowed with a dualizing complex (1.7). The family of functors (cf. (4.5.3))

$$\mathrm{H}_{\bar{x}}^i : \mathrm{D}_c^b(X) \longrightarrow (A\text{-modules})$$

(i running through $\mathbb{Z}$ and $\bar{x}$ through the set of geometric points of X) is “conservative”, i.e. a morphism $F \rightarrow G$ in $\mathrm{D}_c^b(X)$ such that the maps $\mathrm{H}_{\bar{x}}^i(F) \rightarrow \mathrm{H}_{\bar{x}}^i(G)$ are isomorphisms, is an isomorphism.

4.6

Let X be a prescheme, $j : Y \rightarrow X$ a closed sub-prescheme, and $i : \bar{x} \rightarrow Y$ a geometric point of Y . Let, on the other hand, K_X be a dualizing complex on X , normalized at $\bar{x}$ (cf. (4.5)). We saw in (4.5) that, if Y is the closure of x , one has, for $F \in \mathrm{ob} \mathrm{D}_c^b(X)$, a perfect pairing

$$\mathrm{R}^! j(F)_{\bar{x}} \times \underline{D}_X(F)_{\bar{x}} \longrightarrow A.$$

It is not difficult to define an analogous pairing in the general case. Introduce the factorization of i as

$$\bar{x} \xrightarrow{i'} \{\bar{x}\} \xrightarrow{i''} Y.$$

For $F \in \mathrm{ob} \mathrm{D}_c^b(X)$, one has canonical isomorphisms:

$$\begin{aligned} \mathrm{R}\Gamma_{\bar{x}} j^* \underline{D}_X(F) &\xrightarrow{\sim} \mathrm{R}\Gamma_{\bar{x}} \underline{D}_Y \mathrm{R}^! j(F) && \text{(complementary induction formula (4.2.1))} \\ &\xrightarrow{\sim} i'^* \mathrm{R}^! i''^* \underline{D}_Y \mathrm{R}^! j(F) && \text{(definition of } \mathrm{R}\Gamma_{\bar{x}} \text{ (4.5.1))} \\ &\xrightarrow{\sim} i'^* \underline{D}_{\bar{x}} i''^* \mathrm{R}^! j(F) && \text{(ordinary induction (1.12 b) (i))} \\ &\xrightarrow{\sim} \underline{D}_{\bar{x}} i^* \mathrm{R}^! j(F) && \text{(by 4.3)), i.e. finally a} \end{aligned}$$

canonical isomorphism

$$(4.6.1) \quad \mathrm{R}^! j(F)_{\bar{x}} \xrightarrow{\sim} \underline{D}_{\bar{x}} \mathrm{R}\Gamma_{\bar{x}} j^* \underline{D}_X(F),$$

and hence perfect pairings

$$(4.6.1)' \quad \mathrm{R}^! j(F)_{\bar{x}} \times \mathrm{R}\Gamma_{\bar{x}} j^* \underline{D}_X(F) \longrightarrow A,$$

$$(4.6.1)'' \quad \mathrm{H}_Y^i(F)_{\bar{x}} \times \mathrm{H}_{\bar{x}}^{-i}(\underline{D}_X(F)|Y) \longrightarrow A.$$

Example 4.6.2. Let X be a regular prescheme, let $\bar{x}$ be a geometric point of X , and put $d = \dim(\mathcal{O}_{X, \bar{x}})$. Set $K_X = (\mu_n)_X^{\otimes d}[2d]$. If X satisfies condition (reg) (3.1.1) and the purity theorem (3.1.4), then K_X is canonically normalized at $\bar{x}$. Thus, if K_X is dualizing, one has perfect pairings

$$\mathrm{H}_Y^i(F)_{\bar{x}} \times \mathrm{H}_{\bar{x}}^{2d-i}(F'|Y) \longrightarrow A,$$

for every closed subset Y containing x and every $F \in \text{ob } D_c^b(X)$, with $F' = R \text{Hom}(F, (\mu_n)_X^{\otimes d})$.

To finish, let us give a strictly local variant of (4.6.1).

4.7

Let X be a noetherian strictly local scheme (SGA VIII 4.2), with closed point x . Let Y be a non-empty closed subset of X , and consider the canonical immersions

$$U = X - Y \xrightarrow{i} V = X - \{x\} \xrightarrow{j} X.$$

Finally, let K_X be a dualizing complex normalized at x (4.1), giving dualizing complexes “corresponding” on U, V, Y, x (and by hypothesis $\underline{D}_x = R \text{Hom}(_, A)$):

$$K_V = j^* K_X, \quad K_U = i^* K_V, \quad K_Y = R\Gamma_Y(K_X).$$

Let $G \in \text{ob } D_c^b(U)$, and put $G' = \underline{D}_U(G)$. We propose to define a natural pairing

$$(4.7.1) \quad R\Gamma(U, G) \times R\Gamma(V, i_! G')[-1] \longrightarrow A,$$

i.e. a functorial homomorphism

$$(4.7.1)' \quad R\Gamma(U, G) \overset{\mathbf{L}}{\otimes}_A R\Gamma(V, i_! G')[-1] \longrightarrow A.$$

(Since the ring $A = \mathbb{Z}/\ell\mathbb{Z}$ has infinite cohomological dimension for $\ell \geq 2$, one knows how to define only the tensor products

$$\overset{\mathbf{L}}{\otimes} : D^-(A) \times D^-(A) \longrightarrow D^-(A)$$

and

$$\overset{\mathbf{L}}{\otimes} : D(A) \times D^b(A)_{\text{torf}} \longrightarrow D(A) \quad (*).$$

We therefore leave it to the reader to make suitable hypotheses ensuring that the two factors of $\overset{\mathbf{L}}{\otimes}$ appearing in (4.7.1)' are in $D^b(A)$, or that one of them has finite tor-dimension.)

First note that, by the complementary duality formula (1.12 a) (ii), one has a canonical isomorphism

$$(4.7.2) \quad i_!(G') \xrightarrow{\sim} \underline{D}_V(Ri_* G).$$

On the other hand, one has the ordinary duality isomorphism

$$(4.7.3) \quad R \text{Hom}(A_U, G) \xrightarrow{\sim} R \text{Hom}(A_V, Ri_* G).$$

By the definition of $\underline{D}_V$ (and independently of the fact that K_V is dualizing), one has a canonical arrow

$$(4.7.4) \quad Ri_* G \overset{\mathbf{L}}{\otimes} \underline{D}_V(Ri_* G) \longrightarrow K_V.$$

From (4.7.2), (4.7.3), and (4.7.4) one obtains a (canonical) arrow

$$(4.7.5) \quad R\Gamma(U, G) \overset{\mathbf{L}}{\otimes}_A R\Gamma(V, i_! G') \longrightarrow R \text{Hom}(A_V, K_V).$$

But $K_V = j^* K_X$, and one has the duality isomorphism (SGA 3.1.10):

$$(4.7.6) \quad \mathrm{R} \mathrm{Hom}(A_V, K_V) \xrightarrow{\sim} \mathrm{R} \mathrm{Hom}(A_{V,X}, K_X).$$

Finally, the exact sequence

$$(4.7.7) \quad 0 \longrightarrow A_{V,X} \longrightarrow A_X \longrightarrow A_x \longrightarrow 0$$

gives a homomorphism of degree +1:

$$(4.7.8) \quad \mathrm{R} \mathrm{Hom}(A_{V,X}, K_X) \longrightarrow \mathrm{R} \mathrm{Hom}(A_x, K_X) \xrightarrow{\sim} A.$$

Composing (4.7.5), (4.7.6), and (4.7.8), one obtains the announced arrow (4.7.1)'.
() (added in 1977) Inexact: in fact it is easy to extend $\overset{\mathbf{L}}{\otimes} : \mathrm{D}^-(A) \times \mathrm{D}^-(A) \rightarrow \mathrm{D}^-(A)$ to $\overset{\mathbf{L}}{\otimes} : \mathrm{D}^-(A) \times \mathrm{D}(A) \rightarrow \mathrm{D}(A)$.*

We shall now see that the pairing (4.7.1) is “perfect”, i.e. that the arrow (4.7.1)' identifies $\mathrm{R}\Gamma(V, i_! G')[-1]$ with $\underline{D}_x \mathrm{R}\Gamma(U, G)$. Indeed suppose, as is allowed, that

$$G = (ji)^* F,$$

with $F \in \mathrm{ob} \mathrm{D}_c^b(X)$. Put $F' = \underline{D}_X(F)$ and $F'_{U,X} = (ji)_! G'$. One has the exact sequence

$$(4.7.9) \quad 0 \longrightarrow F'_{U,X} \longrightarrow F' \longrightarrow F'|Y \longrightarrow 0,$$

which gives rise to a distinguished triangle

$$\begin{array}{ccc} & \mathrm{R}\Gamma_x(F'_{U,X}) & \\ \swarrow & & \nwarrow +1 \\ \mathrm{R}\Gamma_x(F'|Y) & \xrightarrow{\quad\quad\quad} & \mathrm{R}\Gamma_x(F') \end{array} \quad (b)$$

By definition of $\mathrm{R}\Gamma_x$, one has

$$\mathrm{R}\Gamma_x(F'_{U,X}) = \mathrm{R} \mathrm{Hom}(A_x, F'_{U,X});$$

hence, using (4.7.7) and the fact that $(F'_{U,X})_x = \mathrm{R}\Gamma(X, F'_{U,X}) = 0$, one has a canonical isomorphism

$$\mathrm{R} \mathrm{Hom}(A_{V,X}, F'_{U,X})[1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X}).$$

But one has

$$\begin{aligned} \mathrm{R} \mathrm{Hom}(A_{V,X}, F'_{U,X}) &= \mathrm{R} \mathrm{Hom}(j_! A_V, j_! i_! G') \\ &\xrightarrow{\sim} \mathrm{R} \mathrm{Hom}(A_V, i_! G') \quad \text{by duality (SGA 3.1.10),} \end{aligned}$$

and hence finally a canonical isomorphism

$$(4.7.10) \quad \mathrm{R}\Gamma(V, i_! G')[-1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X}).$$

On the other hand one has the exact sequence

$$(4.7.11) \quad 0 \longrightarrow A_{U,X} \longrightarrow A_X \longrightarrow A_Y \longrightarrow 0,$$

which gives rise to a distinguished triangle

$$\begin{array}{ccc}
 & \mathrm{R}\Gamma(U, F|U) & \\
 \swarrow & & \nwarrow^{+1} \\
 \mathrm{R}\Gamma_Y(F) & \xrightarrow{\quad\quad\quad} & \mathrm{R}\Gamma_X(F)
 \end{array} \quad (a)$$

By virtue of (4.7.1) and (4.7.10), one has a natural pairing

$$(4.7.12) \quad \mathrm{R}\Gamma(U, F|U) \times \mathrm{R}\Gamma_x(F'_{U,X}) \longrightarrow A.$$

By virtue of (4.6.1), one has a perfect pairing

$$(4.7.13) \quad \mathrm{R}\Gamma_Y(F) \times \mathrm{R}\Gamma_x(F'|Y) \longrightarrow A.$$

Finally, by the ordinary induction formula (1.12 b) (i)), one has a perfect pairing

$$(4.7.14) \quad \mathrm{R}\Gamma(X, F) \times \mathrm{R}\Gamma_x(F') \longrightarrow A.$$

The reader who has the patience to orient himself in this maze of “canonical arrows” will check that the pairings (4.7.12), (4.7.13), and (4.7.14) are “compatible” with the arrows of the triangles (a) and (b), and consequently that the pairings (4.7.12) and (4.7.1) are perfect.

The pairing of the triangles (a) and (b) gives rise to two exact sequences “transpose to one another”:

$$\begin{aligned}
 (4.7.15) \quad & \cdots \longrightarrow \mathrm{H}_Y^i(X, F) \longrightarrow \mathrm{H}^i(X, F) \longrightarrow \mathrm{H}^i(U, F) \longrightarrow \cdots \\
 & \cdots \longleftarrow \mathrm{H}_x^{-i}(Y, F'|Y) \longleftarrow \mathrm{H}_x^{-i}(X, F') \longleftarrow \mathrm{H}_x^{-i-1}(V, F'_{U,V}) \longleftarrow \cdots
 \end{aligned}$$

In summary, we may state:

Proposition 4.7.16. Let X be a noetherian strictly local scheme, with closed point x . Let Y be a non-empty closed subset of X , let $U = X - Y$, $V = X - x$, and let $i : U \rightarrow V$ and $j : V \rightarrow X$ be the canonical inclusions. Let K_X be a dualizing complex on X normalized at x (hence giving associated dualizing complexes on V, U, Y). For $G \in \mathrm{ob} \mathrm{D}_c^b(U)$, one has a perfect pairing:

$$(*) \quad \mathrm{R}\Gamma(U, G) \times \mathrm{R}\Gamma(V, i_! G')[-1] \longrightarrow A$$

(where $G' = \underline{D}_U(G)$). If $G = F|U$, where $F \in \mathrm{ob} \mathrm{D}_c^b(X)$, one has a canonical isomorphism

$$(**) \quad \mathrm{R}\Gamma(V, i_! G')[-1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X})$$

(where $F' = \underline{D}_X(F)$ and $F'_{U,X} = (ji)_!(F'|U) = (ji)_!(G')$), and a perfect pairing between the distinguished triangles

$$\begin{aligned}
 & \mathrm{R}\Gamma_Y(F) \longrightarrow \mathrm{R}\Gamma(X, F) \longrightarrow \mathrm{R}\Gamma(U, F|U) \xrightarrow{+1}, \\
 & \mathrm{R}\Gamma_x(F'_{U,X}) \longrightarrow \mathrm{R}\Gamma_x(F') \longrightarrow \mathrm{R}\Gamma_x(F'|Y) \xrightarrow{+1}.
 \end{aligned}$$

compatible with the pairing (*) and the isomorphism (**), and giving rise to two exact sequences transpose to one another (4.7.15).

Remark 4.7.17. In the particular case where $Y = \{x\}$, the pairing (4.7.1) is defined under the sole hypothesis that K_X is normalized at x (and not necessarily dualizing), i.e. such that $\mathrm{R}\Gamma_x(K_X) \simeq A$

(the isomorphism being given): the isomorphism (4.7.2) (which used biduality) then reduces to the identity. Since the pairing (4.7.14) is in any case perfect, one sees therefore that local duality on X (in the form (4.2.2)') is equivalent to the "global" duality on $U = X - x$, i.e. to the assertion that the pairing (4.7.1)

$$R\Gamma(U, G) \times R\Gamma(U, G')[-1] \longrightarrow A$$

is perfect, or equivalently that the pairings

$$H^i(U, G) \times H^{-i-1}(U, G') \longrightarrow A$$

are perfect.

Suppose that one has on X a dualizing complex K_X normalized at x . If X is excellent and U is regular of dimension d (x may therefore be an isolated singularity), one should be able to check that $K_U = K_X|_U$ is canonically isomorphic to $(\mu_n)_U^{\otimes d}[2d]$ (this is easily checked, at any rate, if X is the strict localization of a prescheme locally of finite type over a field of characteristic 0). Then local duality on X implies that, for every locally constant constructible A_U -module G , one has perfect pairings:

$$H^i(U, G) \times H^{2d-1-i}(U, G^\circ) \longrightarrow A,$$

where $G^\circ = \text{Hom}(G, (\mu_n)_U^{\otimes d})$, corresponding formally to Poincaré duality on a scheme of dimension $2d - 1$.

5. Local duality on curves

Theorem 5.1. Let X be a noetherian regular prescheme of dimension 1. Suppose that n is prime to the residual characteristics of X . Then the purity theorem is true on X (3.1.4), and the complex A_X is dualizing (1.7).

Proof. Suppose purity has been proved. Since X plainly satisfies condition (reg) (3.1.1), and since condition (cdloc) $_n$ is verified by (SGA X 2.2), it follows from (3.2.1) that $\dim .q.inj(A_X) = 2$. On the other hand, condition (ii) of (1.7) is verified by (3.3.1 b) (i)) (which was proved assuming purity). Hence, by (4.4), we are reduced to proving that A_X is dualizing when X is strictly local. But, to prove purity, which is a local question in a neighborhood of a closed point, one may equally well suppose X strictly local. Finally, we are reduced to proving the theorem when X is strictly local.

Thus suppose that X is strictly local. Let x be its closed point, let $(\pi = \text{car}(k(x)))$, let η be its generic point, and let $U = X - x = \text{Spec}(k(\eta))$. Let $G = \text{Gal}(k(\bar{\eta})/k(\eta))$ be the fundamental group of U . It is known [CL, Chap. IV § 2 Exer. 2] that one has a canonical exact sequence

$$(5.1.1) \quad 1 \longrightarrow P \longrightarrow G \longrightarrow H \longrightarrow 1,$$

where $H = \prod_{\ell \neq \pi} \mathbb{Z}_\ell(1)$, $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^n}$, and P is a pro- π -group. (Of course, H is isomorphic, but not canonically, to $\prod_{\ell \neq \pi} \mathbb{Z}_\ell$.)

It is also known (SGA VIII 2) that the étale cohomology of U is interpreted as the Galois cohomology of G , sheaves (respectively constructible sheaves) of A_U -modules corresponding to A -modules (respectively finite type A -modules) on which G acts continuously.

This being so, the $H_x^i(A_X)$ are determined by the exact sequence

$$(5.1.2) \quad 0 \longrightarrow H_x^0(A_X) \longrightarrow H^0(X, A_X) \longrightarrow H^0(U, A_U) \longrightarrow H_x^1(A_X) \longrightarrow 0,$$

and the isomorphisms

$$(5.1.3) \quad H_x^i(A_X) \xrightarrow{\sim} H^{i-1}(U, A_U) \quad \text{for } i \geq 2.$$

But trivially $H^0(X, A_X) = H^0(U, A_U) = A$, so $H_x^0(A_X) = H_x^1(A_X) = 0$.

On the other hand, for a Galois G - A -module M , one has the Hochschild–Serre spectral sequence

$$E_2^{pq} = H^p(H, H^q(P, M)) \implies H^{p+q}(G, M).$$

Since n is prime to π and P is a pro- π -group, one has

$$H^q(P, M) = 0 \quad \text{for } q \neq 0,$$

hence $E_2^{pq} = 0$ for $q \neq 0$, giving a canonical isomorphism

$$(5.1.4) \quad H^p(G, M) \xrightarrow{\sim} H^p(H, M^P).$$

By an analogous argument, one sees moreover that $H^p(H, N)$ is identified with $H^p(\widehat{\mathbb{Z}}(1), N)$ when the factor $\widehat{\mathbb{Z}}(1)$ is made to act trivially on N . Thus, finally, with the preceding convention, one has a canonical isomorphism

$$(5.1.5) \quad H^p(G, M) \xrightarrow{\sim} H^p(\widehat{\mathbb{Z}}(1), M^P).$$

Now the Galois cohomology of $\widehat{\mathbb{Z}}$ is well known (CG chap. I p. 31): for a G - A -module N , one has

$$(5.1.6) \quad H^i(\widehat{\mathbb{Z}}, N) = 0 \quad \text{for } i \geq 2, \quad H^0(\widehat{\mathbb{Z}}, N) = N^{\widehat{\mathbb{Z}}}, \quad H^1(\widehat{\mathbb{Z}}, N) = N_{\widehat{\mathbb{Z}}}.$$

From (5.1.3), (5.1.5), and (5.1.6) one immediately deduces that

$$H^i(U, A_U) = 0 \quad \text{for } i \geq 2,$$

and

$$H^1(U, A_U) \xrightarrow{\sim} A.$$

This last isomorphism is not canonical, since it depends on the choice of an identification of $\widehat{\mathbb{Z}}(1)$ with $\widehat{\mathbb{Z}}$. But, using the Kummer exact sequence, one finds (SGA XIX 1.3) a canonical isomorphism (given by the “local fundamental class”)

$$(5.1.7) \quad H^1(U, \mu_n) \xrightarrow{\sim} A.$$

Purity is therefore proved.

Let us now prove that A_X is dualizing. By (4.7.17), it is a matter of showing that, for every locally constant constructible A_U -module M , the pairing

$$(5.1.8) \quad H^i(U, M) \times H^{1-i}(U, \text{Hom}(M, A_U)) \longrightarrow H^1(U, A_U) \xrightarrow{\sim} (\mu_n)_x^{\otimes -1}$$

is a perfect duality. We shall interpret this pairing in terms of Galois cohomology. We shall use the following elementary lemma.

Lemma 5.1.9. Let $D = \text{Hom}(_, A)$ denote the Pontrjagin dualizing functor on the category of finite type A -modules. Let M be a finite type A -module on which a group G acts A -linearly.

(i) One has a canonical isomorphism

$$D(M^G) \xrightarrow{\sim} (DM)_G.$$

(ii) If G is a profinite group acting continuously on M , and if G has pro-order prime to n , then the canonical arrow

$$M^G \longrightarrow M_G$$

is an isomorphism.

Proof. By definition one has

$$M^G = \varprojlim_{s \in G} \text{Ker}(M \xrightarrow{s-1} M),$$

hence

$$D(M^G) \xrightarrow{\sim} \varprojlim_{s \in G} \text{Coker}(DM \xrightarrow{s-1} DM) \xrightarrow{\sim} (DM)_G,$$

which proves (i).

Let us prove (ii). Let M' be the A -module defined by the exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M_G \longrightarrow 0.$$

One has $H^1(G, M') = 0$ because G has pro-order prime to n . The exact cohomology sequence therefore implies that the arrow $M^G \rightarrow M_G$ is an epimorphism. But, applying the functor D and taking (i) into account, one deduces that it is a monomorphism, which completes the proof.

Applying (5.1.9) to M and P , one finds that $\text{Hom}(M, A)^P$ is canonically identified with $\text{Hom}(M^P, A)$. If one chooses an isomorphism from $\widehat{\mathbb{Z}}(1)$ to $\widehat{\mathbb{Z}}$, the pairing (5.1.8) is written, taking (5.1.5) into account, as

$$(5.1.8)' \quad H^i(\widehat{\mathbb{Z}}, M^P) \times H^{1-i}(\widehat{\mathbb{Z}}, D(M^P)) \longrightarrow H^1(\widehat{\mathbb{Z}}, A) \xrightarrow{\sim} A.$$

It follows immediately from (5.1.6) and (5.1.9(i)) that (5.1.8)' is a perfect pairing. This completes the proof of (5.1).

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